Does Bitcoin Behave Like a Risky Financial Asset?

Financial Econometrics Exam Project

Sample: 2017-01-04 to 2026-05-21; 2341 daily business-day observations.

R code and downloaded data are supplied separately for reproducibility.

1. Introduction and Background

Bitcoin has developed from a niche payment technology into a large traded financial asset. A central empirical question is whether its return dynamics resemble those of a risky speculative asset: high volatility, heavy tails, volatility clustering, and short-run co-movement with broader financial markets. This project asks whether Bitcoin behaves like a risky financial asset, and whether equity-market, uncertainty, and interest-rate shocks help explain Bitcoin returns.

The question matters because Bitcoin is often discussed both as an alternative currency and as a portfolio asset. Previous studies have debated whether Bitcoin behaves mainly as a medium of exchange, speculative asset, or diversifier. This project keeps the literature review brief and focuses on applying time-series econometric tools from the course.

The hypotheses are: (1) Bitcoin returns have weak predictable mean dynamics but clear conditional heteroskedasticity; (2) Bitcoin is contemporaneously related to equity-market and uncertainty conditions; and (3) lagged equity, VIX, and interest-rate changes have limited predictive power for Bitcoin returns.

2. Data

The data are daily observations downloaded from FRED, Federal Reserve Bank of St. Louis. The core series are Coinbase Bitcoin prices in U.S. dollars (CBBTCUSD), the S&P 500 index (SP500), the CBOE VIX index (VIXCLS), and the 10-year U.S. Treasury yield (DGS10). Bitcoin trades every day, while the other variables are business-day series. The empirical sample therefore keeps common dates where all core variables are observed.

The model variables are Bitcoin and S&P 500 daily log returns, r_t = 100 * Delta log(P_t), together with first differences in VIX and the 10-year yield. This transformation is motivated by the usual non-stationarity of financial price levels and by the stationarity requirements of ARIMA, GARCH, and VAR models.

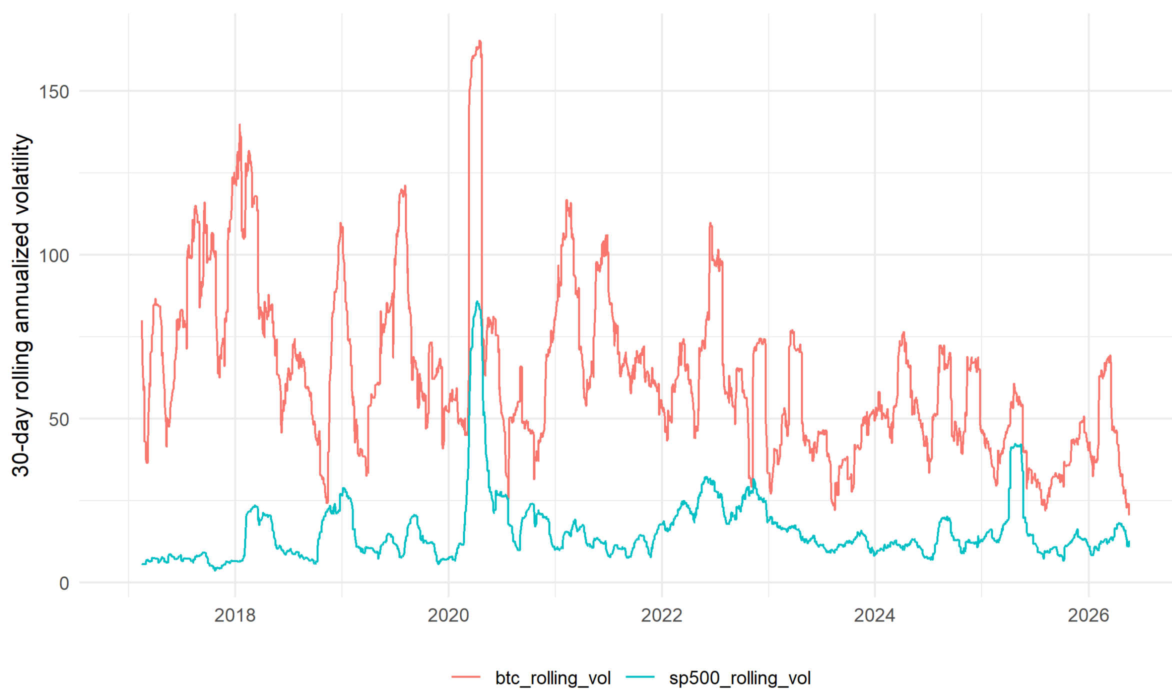
Table 1. Summary statistics

Variable	N	Mean	SD	Min	Max	Skew.	Excess kurt.
btc_return	2341	0.185	4.329	-46.862	24.061	-0.544	9.41
sp500_return	2341	0.051	1.169	-12.765	9.089	-0.685	16.25
vix_change	2341	0.002	2.018	-18.71	24.86	2.14	31.905
dgs10_change	2341	9.06e-04	0.054	-0.3	0.29	-0.073	2.032

Bitcoin returns are much more volatile than S&P 500 returns. The daily standard deviation of Bitcoin returns is 4.329 percent, compared with 1.169 percent for the S&P 500. Bitcoin also displays negative skewness and high excess kurtosis, consistent with large downside movements and fat tails.

Figure 1. Indexed Bitcoin, S&P 500, and Nasdaq prices

Figure 2. Rolling annualized volatility



3. Methodology

The first model family is ARIMA/GARCH. I model Bitcoin returns with an ARIMA process, phi(L)(r_t - mu) = theta(L)e_t, where phi(L) and theta(L) are autoregressive and moving-average lag polynomials. The ARIMA model describes the conditional mean and provides a simple forecasting benchmark.

Because financial returns often have time-varying variance, I then estimate a GARCH(1,1) model: e_t = sigma_t z_t and sigma_t^2 = omega + alpha e_(t-1)^2 + beta sigma_(t-1)^2. The alpha parameter measures the immediate effect of shocks on volatility, while beta measures volatility persistence.

The second model family is a vector autoregression. The VAR contains S&P 500 returns, VIX changes, 10-year Treasury yield changes, and Bitcoin returns: $y_t = c + A_1 y_{t-1} + u_t$. The lag length is selected by information criteria. Granger causality tests evaluate whether lagged traditional financial variables improve prediction of Bitcoin returns, and impulse response functions trace the reaction of Bitcoin returns to market shocks.

4. Results

Table 2. Unit-root test p-values

Variable	ADF p-value	KPSS p-value
log_btc_price	0.193	0.01
log_sp500	0.145	0.01
dgs10	0.778	0.01
btc_return	0.01	0.1
sp500_return	0.01	0.1
vix_change	0.01	0.1
dgs10_change	0.01	0.1

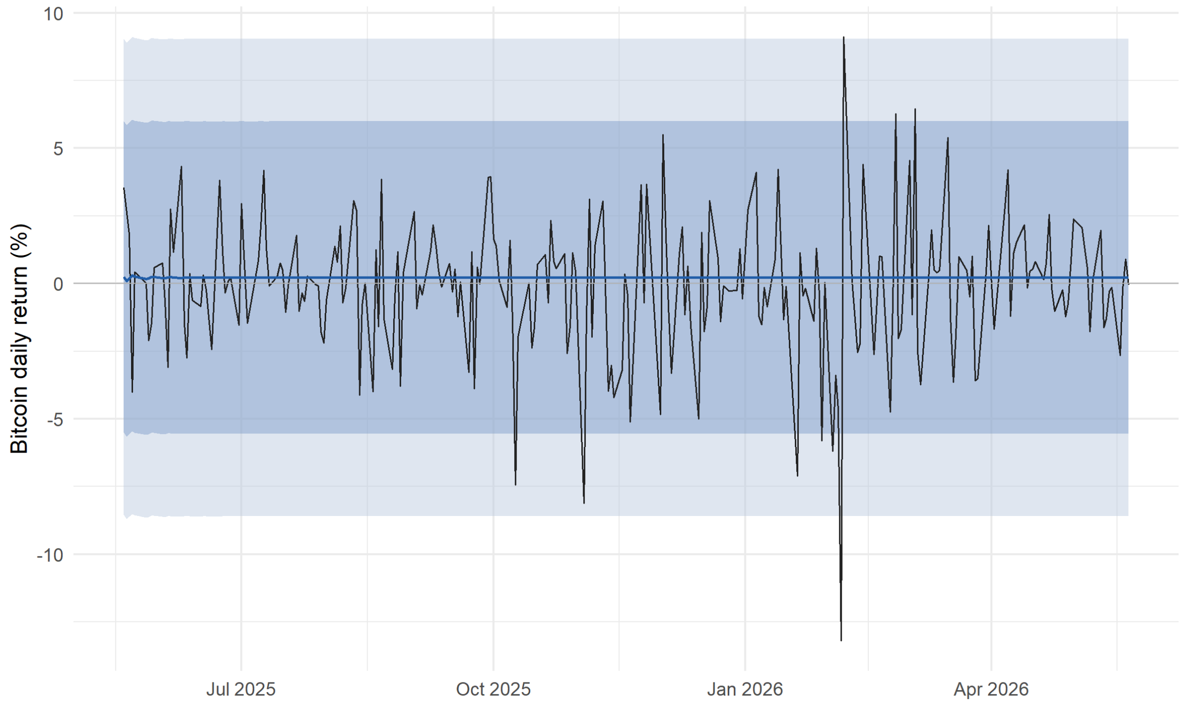
The unit-root tests support transforming the price-level variables. For log Bitcoin and log S&P 500 prices, ADF tests do not reject a unit root, while KPSS tests reject level stationarity. For the transformed variables, ADF tests reject a unit root and KPSS tests do not reject stationarity.

Table 3. ARIMA and GARCH results

Output	Value
Selected ARIMA model	ARIMA(2,0,2) with non-zero mean
Ljung-Box p-value	0.754
ARCH test p-value	5.38e-04
Test RMSE	2.551
GARCH alpha	0.102
GARCH beta	0.831
alpha + beta	0.933

Automatic ARIMA selection chooses an ARIMA(2,0,2) model with non-zero mean. The Ljung-Box p-value of 0.754 indicates no strong remaining residual autocorrelation, but the ARCH test p-value of 5.38e-04 indicates conditional heteroskedasticity. The GARCH coefficients are positive and the alpha plus beta sum is approximately 0.933, implying persistent volatility.

Figure 3. ARIMA forecasts for the test sample



The conditional mean forecasts remain close to zero and realized returns are mostly unpredictable within wide forecast intervals. This supports the view that Bitcoin's mean return is hard to forecast, even though its volatility is highly time-varying.

5. VAR, Granger Causality, and Impulse Responses

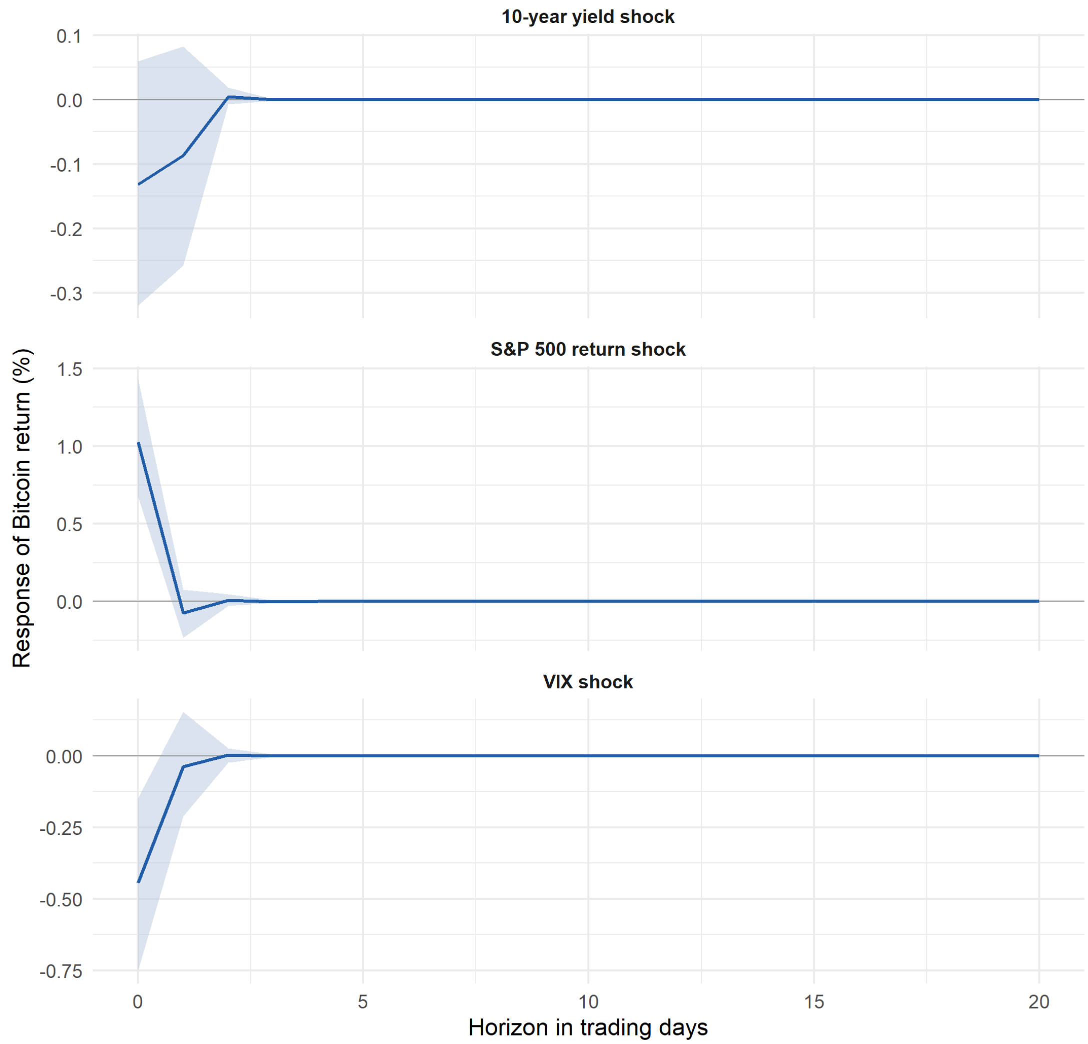
The Schwarz criterion selects 1 lag for the VAR. Bitcoin has a positive contemporaneous correlation with S&P 500 returns of about 0.236 and a negative correlation with VIX changes of about -0.247. This is more consistent with risky-asset behavior than with safe-haven behavior.

Table 4. Granger causality tests for the Bitcoin equation

Null hypothesis	Lags	F-stat.	p-value
sp500_return does not Granger-cause btc_return	1	0.49	0.484
vix_change does not Granger-cause btc_return	1	0.46	0.498
dgs10_change does not Granger-cause btc_return	1	1.061	0.303

In the Bitcoin equation, none of the lagged traditional financial variables significantly Granger-causes Bitcoin returns at conventional levels. The p-values for lagged S&P 500 returns, VIX changes, and Treasury yield changes are all above 10 percent. Thus, Bitcoin is connected to market conditions contemporaneously, but these variables do not strongly predict next-day Bitcoin returns.

Figure 4. Bitcoin impulse responses to market shocks



The impulse response functions show an immediate positive response of Bitcoin returns to an S&P 500 shock and an immediate negative response to a VIX shock. These effects decay quickly and are not persistent over later horizons. The response to a Treasury-yield shock is weaker and statistically less clear. Because the impulse responses are orthogonalized, the ordering matters; equity returns and VIX are placed before Bitcoin to interpret them as broader market shocks hitting Bitcoin within the day.

6. Conclusion

The evidence suggests that Bitcoin behaves more like a risky financial asset than a currency or safe haven. Bitcoin returns are far more volatile than S&P 500 returns, have fat tails, and display strong conditional heteroskedasticity. The ARIMA model removes most mean autocorrelation, but the ARCH test and GARCH estimates show that volatility clustering is economically and statistically important.

The VAR results add a second perspective. Bitcoin is positively correlated with equity returns and negatively correlated with VIX changes, suggesting risk-on/risk-off behavior. However, lagged S&P 500 returns, VIX changes, and Treasury yield changes do not significantly Granger-cause Bitcoin returns in the Bitcoin equation. Overall, Bitcoin is best understood as a highly volatile speculative asset with weak predictable mean dynamics and persistent volatility.

A limitation is that the analysis uses a compact VAR and daily data. Future work could compare subsamples, include the dollar index or liquidity variables, or estimate a multivariate volatility model.

References

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